

Rithwik K Kamalesh

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EDUCATION

Georgia Institute of Technology

Atlanta, Georgia

Masters of Science in Computer Science (Part-time)

January 2026 – December 2027

- **Concentration:** Artificial Intelligence/ ML
- **Relevant Coursework:** Machine Learning for Trading (CS 7646), DB Sys Concepts & Design (CS 6400)

Georgia State University

Atlanta, Georgia

Bachelor of Science in Computer Science

August 2021 - December 2024

- **Concentration:** Data Science
- **Relevant Coursework:** Data Structures (DSCI 2720), Probability & Stats for CSC (MATH 3020), Big Data (DSCI 4760), Algorithms (CSC 4520), Data Mining (DSCI 4740), Machine Learning (DSCI 4850), Data Science Capstone (DSCI 4760)
- **Clubs:** Beta Alpha Psi, Financial Technology Organization (FTO), Cybersecurity Club at GSU

RELEVANT WORK EXPERIENCE

Roamio.ai

Atlanta, GA

Data Analyst

May 2025- Present

- Analyzed **10K+ user events** to identify key engagement trends, improving overall retention metrics by **~15%**.
- Built **5+ Power BI dashboards** tracking **5 KPIs** to diagnose low event attendance, presenting insights to stakeholders.
- Partnered with the marketing team to strategically advertise events using dashboard reports, increasing attendance by **~6%**.
- Applied data analysis to improve **UX and feature performance**, boosting overall conversion rates by **~12%**.
- Monitored user behavior and engagement metrics, providing data insights to support product improvements

Big Brother Big Sister of America

Atlanta, GA

Data Analyst Intern

May 2024-November 2024

- Help develop **ML matching system** with **1,000+** synthetic IDs, improving data governance enabling secure analytics.
- Processed 5,000+ interview records using **SQL/Python**, reducing inconsistencies by **~30%** and improving **data quality**.
- Engineered **15+ NLP** personality features, improving match accuracy and overall recommendation system performance.
- Performed **data cleaning, validation, and deduplication on datasets**, improving data reliability for downstream modeling
- Analyzed matching outcomes and user attributes to refine **model features**, improving overall match success and recommendation quality

RELEVANT PROJECT EXPERIENCE

AI-Powered HR Conflict Detection System

- Built an end-to-end **agentic AI system** that analyzes candidate resumes and cross-references employer history against 10,000+ SEC 10-K Exhibit 21 subsidiary records, achieving **92% entity matching** precision in detecting potential conflicts.
- Architected a **4-agent LLM pipeline** using **OpenAI APIs** for structured data extraction and **fuzzy entity matching** across resumes and SEC filings, enabling scalable and accurate conflict detection workflows.
- Built an explainable conflict reporting engine with automated **PDF entity highlighting**, reducing compliance review time by **~75%** while improving transparency and interpretability in conflict detection processes
- Deployed system on **AWS with Docker** and secure storage, processing 100+ comparisons every hour.

Image Classification Project

Personal Project

- Trained deep **CNN** on 60K CIFAR-10 images using **TensorFlow/Keras framework**, achieving 92.9% test accuracy overall
- Improved model generalization with **Conv2D, BatchNorm, and Dropout layers**, increasing validation accuracy by **~8%** overall.
- Evaluated overall model performance using **precision, recall, and F1-score metrics**, achieving 0.93 macro F1-score.

Fully Autonomous Drone

Personal Project

- Built a custom autonomous drone using a DJI kit and Jetson Nano, integrating **GPS, LiDAR, and vision sensors** with OpenCV.
- Developed a proprietary real-time flight control algorithm in **Python/C++**, enabling stable navigation and obstacle avoidance
- Integrated sensors and **control algorithms** to enable stable flight and real-time obstacle avoidance.

TECHNICAL SKILLS/CERTIFICATION

Certifications: AWS AI Practitioner, Databricks Generative AI Fundamentals, AWS Associate Machine Learning Engineer*- (2026)

Skills: JavaScript, Java, Python, Node, DynamoDB, MongoDB, Vue, Angular, SQL, React, C++, Control Systems, Kinematics, State Estimation, TCP/IP, REST APIs, Wireshark, Linux, Docker, Kubernetes, Jira, MagicDraw